

PHYSICS LAB REPORT

STRUCTURE

A Physics lab report should contain the following parts:

- Introduction
- Theory
- Setup
- Results and Analysis
- Discussion

INTRODUCTION

Write a short (!) introduction into the topic: What is the goal of the lab? Why is this phenomenon relevant (e.g. applications in everyday life)?

THEORY

Describe in your own words (!) the quantities and relations relevant for the phenomenon investigated. You are not expected to copy lengthy introductions into the topic or derivations of complicated formulae. This part should also be used to introduce the symbols for the physical quantities used consistently throughout your report.

SETUP

Describe the setup used for your measurements. This usually includes a sketch or a photo of the devices used during the lab session.

RESULTS AND ANALYSIS

This is the main part of the report. It starts with an overview of the data collected during the lab sessions (data tables and/or diagrams). Make sure that everything is clearly labelled (with the symbols introduced earlier) and has proper units. Whenever possible measured quantities should be given with an error (error bars in diagrams).

The analysis of the data answers (at least) the questions from the instruction sheet. Explain your steps in clear sentences. Explain how you determine the error of calculated quantities and write the results with their error correctly rounded in the standard form.

DISCUSSION

Give a short overview of the main results. In particular you should repeat the most important numerical results and discuss their validity (e.g. by comparing to a reference value). Reflect your own work and suggest things that could be improved.